ANISH DOKANIA

anishdokania999@gmail.com • (832) 942-4409 • Houston, TX • linkedin.com/in/anishdokania • github.com/anishdokania

EDUCATION

B.S. Computer Science, Minor in Mathematics — University of Houston | GPA 3.72 / 4.00, Cum Laude, Dean's List Aug 2022 – May 2026

Relevant coursework: Mathematical Intro to Options, Probability, Statistics, Linear Algebra, Discrete Math, Algorithms, Data Science I & II

Postgraduate, Sports Analytics — Barça Innovation Hub (FC Barcelona), remote Expected May 2027

Awards: RowdyHacks 2024 Champion (1st of 90+ teams) · UH Ideathon 2024 — 3rd place, “Best College Campus Solution” (top 6 of 56 teams) · Academic Excellence Scholarship (merit)

EXPERIENCE

Data Science Intern — Allstate Insurance Jun 2024 – Aug 2024

- Delivered a full customer-churn proof of concept that was accepted and taken through to production — ensemble model with calibrated probabilities (AUC 0.84) giving retention teams early warning across 5M+ policyholders.
- Won stakeholder confidence by using SHAP interpretability to explain in business terms why customers were flagged, then turned those findings into validated behavioral features that lifted customer-effort prediction precision by 12%.
- Removed the iteration bottleneck by automating the training and evaluation pipeline, cutting experimentation time by 40%.

Gen AI Intern — Ambit Software, India Jun 2023 – Aug 2023

- Built and integrated a generative-AI tool into client CRM platforms, taking repetitive workflow steps off the hands of enterprise users.
- Replaced a manual support-intake process with a customer ticketing portal delivered on PowerApps with a custom HTML/CSS front end.
- Turned recurring manual data pulls into a scheduled Power Automate and Power BI pipeline, so client feedback reporting ran untouched — and shortened configuration-to-deployment time by automating the validation and form-generation steps slowing every handoff.

PROJECTS

Options Swing-Trading Signal & Backtesting Engine — Python · pandas · NumPy · GitHub Actions /Personal-Sean

- Codified a discretionary options swing-trading playbook (The Options Cartel Blueprint) into testable rules: long calls on high-ADR momentum leaders at undercut-and-reclaim / 8 EMA pullbacks and base breakouts, bought ahead of IV expansion.
- Specified the options risk framework: exit at the chart pivot or a −25% premium hard stop (theta and bid/ask decay), trim ¼ at +25% and move the rest to breakeven, structural swing-high targets, and tiered sizing of 0.5–5% of account per trade.
- Built a daily signal pipeline over 5,100+ U.S. equities — liquidity/volatility gates (ATR14, $-volume, ADR% ≥ 5), point-in-time sector-momentum ranking across 11 SPDR ETFs, and multi-factor setup scoring; an LLM chart review sees only the top ~10 names.
- Wrote an event-driven, point-in-time portfolio backtester (no look-ahead): limit-order fills, gap-aware stops, stop-first resolution of ambiguous bars, 1% fixed-fractional sizing, 10-position cap, and slippage — reusing the live scoring code so research and production never drift.
- Evaluated 850+ simulated trades on expectancy (R), profit factor, t-stat, and drawdown; a sector-alignment filter lifted expectancy from +0.02R to +0.07R (PF 1.09), but at t ≈ 0.7 I treated it as noise, not edge.
- Stress-tested the tight-stop entry: isolated that 59% of trades hinge on unobservable entry-day ordering, bounding expectancy at −0.20R to +0.18R and making intraday data the gate before risking capital.

Governed Multi-Agent Content Operator — TypeScript · Next.js · OpenAI API · Playwright /Personal-xAI-Distributor-Manager-Operator

- Built scoring, drafting, and posting agents that rate every inbound item 1–10 for relevance and risk, then layered two independent controls — pattern screening plus an LLM risk pass — with draft-first defaults and rate limits, so a human approves before anything goes out.
- Logged every action, skip, and error to SQLite for a full audit trail, and added a deterministic offline mode so the pipeline can be demoed to a stakeholder at zero API cost.

Real-Time Sign Language Interpreter — RowdyHacks 2024 · Python · React · YOLOv8 devpost.com/software/sign_interpreter_app

- Built a live ASL-to-text translator in 24 hours with a 3-person team — custom gesture model trained on 5K+ labeled images to 92% accuracy at sub-50 ms inference, behind a web UI usable with no technical background — and won first place at the largest hackathon in Texas, pitching to a judging panel against 90+ teams.

RESEARCH

Undergraduate Student Instrumentation Project (USIP) — University of Houston & NASA Aug 2022 – Aug 2024

- Served as the processing pipeline for the microplastics, radio, and other payload research teams — turning 75K+ raw telemetry datapoints per flight into analysis-ready datasets each team could work from directly.
- Selected for NASA-funded balloon missions (2023 Alaska aurora; 2024 Texas solar eclipse); built Python/MATLAB pipelines and version-controlled workflows that kept results reproducible across every team using the data.

TECHNICAL TOOLKIT

Quant & Statistics: probability, statistical inference & hypothesis testing, time-series analysis, backtesting (point-in-time, look-ahead & survivorship control), options pricing & Greeks (Black-Scholes), position sizing & risk

Languages & Engineering: Python, C++, SQL, TypeScript, JavaScript; FastAPI, Next.js, React, Node.js, PostgreSQL; GitHub Actions (CI/CD); Power Automate, PowerApps, Power BI

Data & ML: pandas, NumPy, scikit-learn, SHAP, PyTorch, OpenCV, YOLOv8

AI / LLM: Anthropic (Claude) & OpenAI APIs, structured JSON outputs, tool use, MCP, RAG, LLM-as-judge, evaluation & guardrails

Domains delivered in: insurance, enterprise CRM, capital markets, aerospace research, sports analytics